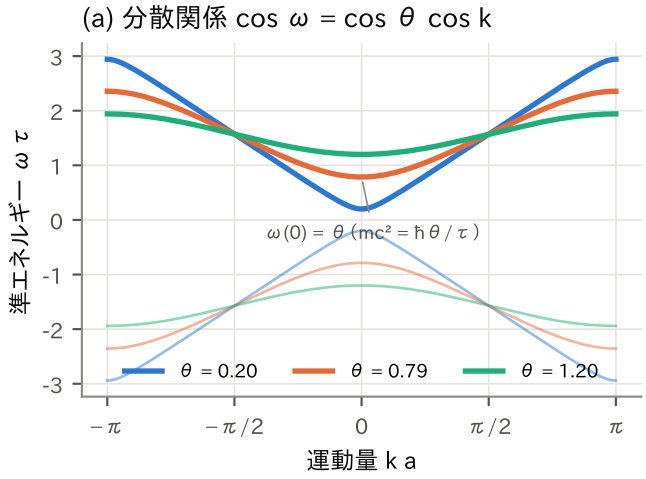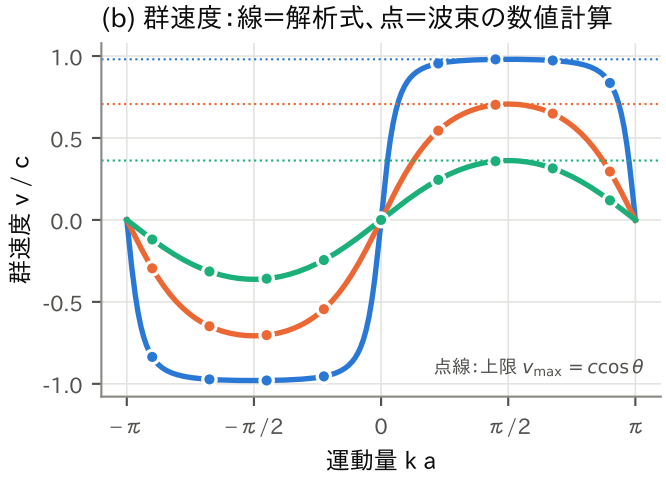図Q2　分散関係と群速度の上限 $v_{\max} = c\cos\theta$